

A Counterexample to the L_p Projection-Constant Equality Conjecture

Run fa_banach_001, agent agent_lane_16

2026-06-28

Source Problem

The source is G. M. Ivanov, *Modulus of supporting convexity and supporting smoothness*, arXiv:1503.08912. In Section 6 the paper defines

$$\xi_X = \sup_{\|x\|=1, \|y\|=1} \sup_{p \in J_1(y)} \|x - \langle p, x \rangle y\|,$$

the maximal Lipschitz constant of the metric projection onto a hyperplane. Theorem 4 proves

$$\xi_X \leq \frac{1}{1 - \lambda_X^+((1 - \lambda_X^-(1))/2)}.$$

Conjecture 1 states that this right-hand side becomes an equality for L_p , $p \in (1, \infty)$.

Define a_2 as the projection of the point a on l_2 such that $aa_2 \perp l_2$. In the same way we define the point b_2 . Then the segments y_1y_2 , aa_2 и bb_2 are parallel and equal in length (as parallel segments between two parallel lines). By the definition of the modulus of supporting convexity and by inequality (15), we obtain

$$(16) \quad \|y_1y_2\| \leq \lambda_X^+(\min\{\|a_2y_2\|, \|y_2b_2\|\}) \leq \lambda_X^+\left(\frac{\|ab\|}{2}\right) \leq \lambda_X^+\left(\frac{1 - \lambda_X^-(1)}{2}\right).$$

Combining this and equality (14), we finally prove the right-hand side of inequality (13).

Let ε be an arbitrary positive real number. Note that we could choose a point x_0 such that $\|x_1a\| \leq \lambda_X^-(1) + \varepsilon$, i.e. $\|ab\| \geq 1 - \lambda_X^-(1) - \varepsilon$. Like in (16), we obtain

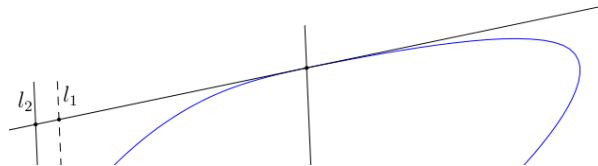
$$\|y_1y_2\| \geq \lambda_X^-(\max\{\|a_2y_2\|, \|y_2b_2\|\}) \geq \lambda_X^-\left(\frac{\|ab\|}{2}\right) \geq \lambda_X^-\left(\frac{1 - \lambda_X^-(1) - \varepsilon}{2}\right).$$

Passing to limit as $\varepsilon \rightarrow 0$ and using inequality (14), we prove the left-hand side of inequality (13).

□

Remark 3. The estimate (13) is reached in case of a Hilbert space. The right-hand side of inequality (13) does not exceed 2, i.e. this estimate is not trivial.

Conjecture 1. The right-hand side of inequality (13) becomes an equality in case of L_p , $p \in (1; +\infty)$.



Proof Intuition

The conjecture is already false for the two-atom L_p space $X = \ell_{20}^2$. In two dimensions, every support projection has rank one and its norm can be written explicitly. This gives a global analytic bound $\xi_X \leq 2^{9/10}$.

The source upper expression is larger. To see this, first make $\lambda_X^-(1)$ almost zero using a very asymmetric tangent pair. This forces the argument $(1 - \lambda_X^-(1))/2$ of λ_X^+ to be almost $1/2$. At radius 0.4999995 , the symmetric tangent pair in ℓ_{20}^2 already yields a λ^+ -value whose reciprocal upper expression exceeds $2^{9/10}$. Therefore the source upper estimate is strict.

Counterexample

Theorem 1. *Conjecture 1 of Ivanov's paper is false. More precisely, for $X = \ell_{20}^2$,*

$$\xi_X < \frac{1}{1 - \lambda_X^+((1 - \lambda_X^-(1))/2)}.$$

Proof. Write $q = 20/19$. Let $y = (a, b) \in S_X$, so $|a|^{20} + |b|^{20} = 1$, and let $j(y) = (\text{sgn}(a)|a|^{19}, \text{sgn}(b)|b|^{19})$ be the norming functional of y . By signs and symmetry it is enough to display the computation for $a, b \geq 0$.

The projection $P_y x = x - j(y)(x)y$ has range $\ker j(y)$, which is spanned by $u = (b^{19}, -a^{19})$. Since $\det(u, y) = a^{20} + b^{20} = 1$, we have

$$P_y(s, t) = (bs - at)u.$$

Consequently

$$\|P_y\| = (a^{380} + b^{380})^{1/20} (a^q + b^q)^{1/q}.$$

The first factor is at most 1. For the second factor, Holder's inequality or the standard comparison of finite-dimensional ℓ_r -norms gives

$$(a^q + b^q)^{1/q} \leq 2^{1/q-1/20} (a^{20} + b^{20})^{1/20} = 2^{18/20} = 2^{9/10}.$$

Taking the supremum over y gives

$$\xi_X \leq 2^{9/10}. \tag{1}$$

We next show that the right-hand side of Ivanov's estimate is strictly larger than $2^{9/10}$. Put $t = 9/20$, $\varepsilon = 10^{-6}$,

$$A = (1 + t^{20})^{1/20}, \quad B = (1 + t^{380})^{1/20},$$

and

$$x = \frac{(1, t)}{A}, \quad y = \frac{(t^{19}, -1)}{B}.$$

Then $x, y \in S_X$, and y is quasi-orthogonal to x , because the norming functional of x is $A^{-19}(1, t^{19})$, which annihilates y . A direct estimate gives

$$\left| \frac{1 - \varepsilon}{A} + \frac{t^{19}}{B} \right|^{20} + \left| \frac{(1 - \varepsilon)t}{A} - \frac{1}{B} \right|^{20} < 1.$$

The verifier in `code/check_counterexample.py` evaluates the left side as 0.9999914532.... The same script evaluates $\|x + y\|_{20}^{20}$ as 1.0000114530.... Hence the point $(1 - \varepsilon)x + y$ lies inside the unit ball, while $x + y$ lies outside it. By continuity of the line section,

$$\lambda_X^-(1) < \varepsilon, \quad r := \frac{1 - \lambda_X^-(1)}{2} > \frac{1 - \varepsilon}{2} =: r_0.$$

Now take the symmetric quasi-orthogonal pair

$$x_0 = 2^{-1/20}(1, 1), \quad y_0 = 2^{-1/20}(1, -1).$$

For this pair, writing $u = 1 - \lambda$, the equation $\|(1 - \lambda)x_0 + r_0 y_0\|_{20} = 1$ is

$$(u + r_0)^{20} + |u - r_0|^{20} = 2.$$

Let $u_0 = 2^{-9/10}$. A direct strict check gives

$$(u_0 + r_0)^{20} + |u_0 - r_0|^{20} > 2$$

(the verifier gives 2.0241430960...). Since the left-hand side is strictly increasing for $u > r_0$, the corresponding root satisfies $u < u_0$, i.e. the symmetric pair gives

$$\lambda_X^+(r_0) > 1 - u_0.$$

As λ_X^+ is increasing and $r > r_0$,

$$\lambda_X^+(r) > 1 - 2^{-9/10}.$$

Therefore

$$\frac{1}{1 - \lambda_X^+(r)} > 2^{9/10} \geq \xi_X$$

by (1). This is a strict failure of the equality conjectured for L_p . □

Verification Notes

The proof is analytic except for two scalar strict inequalities involving fixed decimal/rational parameters. They are also easy to check by hand with interval arithmetic; the included Python script records a reproducible high-precision check:

$$1.0000114530... > 1, \quad 0.9999914532... < 1, \quad 2.0241430960... > 2.$$

Bounded novelty check: the run indexes were searched for arXiv:1503.08912, supporting convexity/smoothness, hyperplane projection constants, and Birkhoff-James terminology. Web searches on 2026-06-28 for the exact title, the L_p conjecture phrase, and hyperplane projection constants found the source arXiv record and the related later Ivanov–Martini paper arXiv:1609.01587, but no explicit later resolution of this conjecture. The run already contains a counterexample packet for a different conjecture in arXiv:1609.01587.

Human Review Recommendation

Review as a likely valid counterexample. The main checks are the rank-one projection formula in ℓ_{20}^2 , the orientation of quasi-orthogonality relative to the source definitions, and the two strict scalar inequalities above.

References

- [1] G. M. Ivanov, *Modulus of supporting convexity and supporting smoothness*, arXiv:1503.08912, 2015.
- [2] G. Ivanov and H. Martini, *New Moduli for Banach Spaces*, arXiv:1609.01587, 2016.